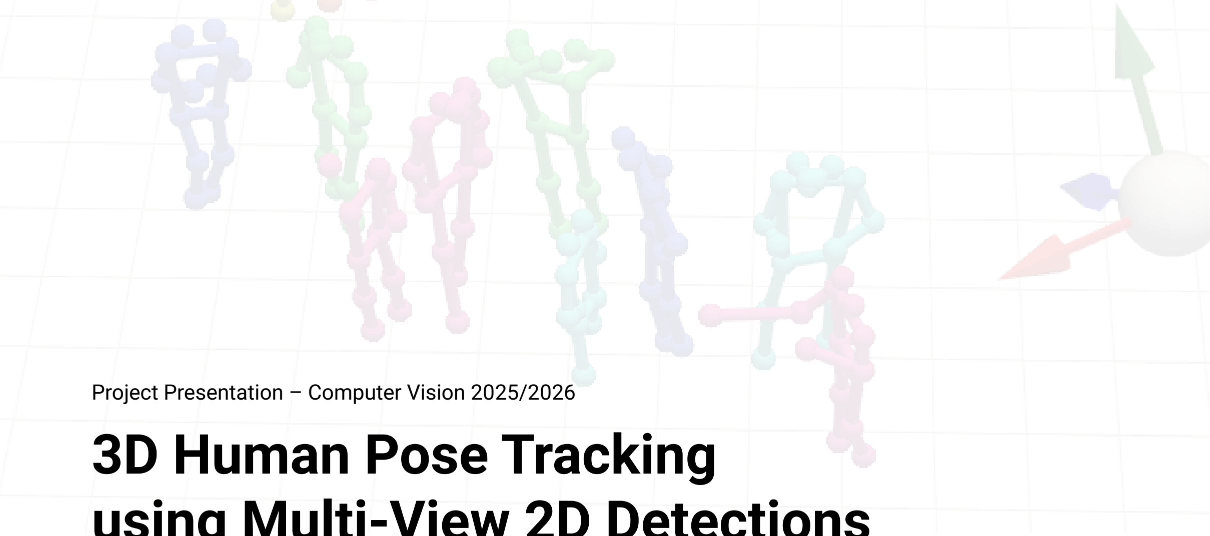


Project Presentation – Computer Vision 2025/2026

3D Human Pose Tracking using Multi-View 2D Detections

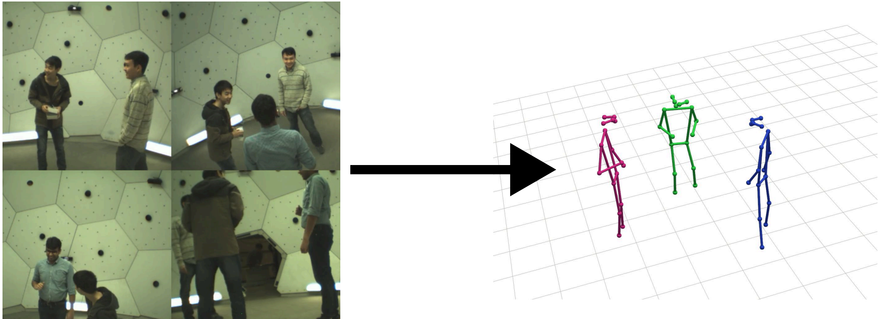
Roland Bernard

June 2026

Task – Human Pose Estimation

- Given:
  - → Multiple video sequences containing multiple people.
  - → Recorded using multiple calibrated cameras.

- Reconstruct:
  - → 3D coordinates of skeletal joint for each subject.
  - → Movement of skeletal joint over time.



How am I trying to solve it?

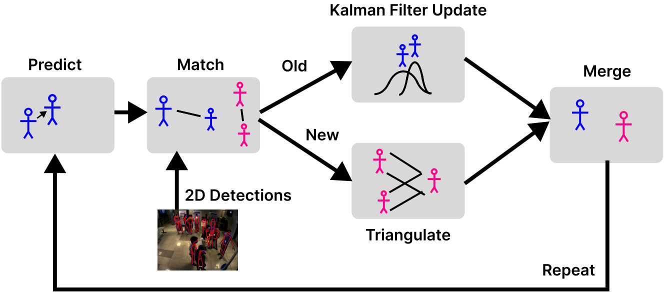
- Summary:
  - → Perform single-view single-frame 2D estimation using pretrained model.
  - → Reconstruct 3D geometry from multiple camera views via triangulation.
  - → Apply Kalman filtering to smooth out noise.

What makes this difficult?

- Challenges:
  - → Occlusions can make some joints unobserved.
  - → Matching predictions between multiple views.
  - → Matching predictions across time.
  - → Noisy 2D estimates make the above more difficult.
  - → 2D pose estimator often misses some people.

Current Approach – Extended Kalman Filter

- Summary:
  → Keep internal 3D state of all currently tracked persons.
  Repeat for each frame:
  → Predict next frame state using current state.
  → Match predictions with detections.
  → Update internal 3D state using detections.
  → Associate unmatched detections between views.
  → Create new internal state using triangulation.



What to do next?

- Learn filter parameters from data.
  → Currently, just guessed manually.

- Add constraint that forces consistency of joint lengths.
  → To improve results if person is partially occluded.

- Try using a particle filter with occlusion aware likelihood.
  → More expressive than Kalman filter.
  → Could accurately model occlusions.